# BrightLearn Data Analytics

## Research Assignment 1

### Foundations of Data Analytics & Data Science

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| Course | Data Analytics & Data Science |
| Assignment No. | Research Assignment 1 |
| Target Audience | Beginner Students |
| Due Date | To be confirmed by instructor |
| Format | Handwritten answers — scan and email your completed work |

## Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

## Section A: Data Fundamentals

**Research Tip:** *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A data base is essentially an organized collection that can be easily accessed managed, and updated. Database are essential today they allow massive amount of data to be stored and retrieved



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quickly, which is critical in a world where information grows every seconds. Database reduces errors by keeping data consistent and centralized. Examples

*Healthcare systems: Patient records, medical histories and prescriptions are stored in database so doctors can access

* Online stores like Amazon or Takealot use database to track product, manage customer orders and recommend items base on browsing history.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse? Data warehouse is a large centralized system designed specifically for storing and analyzing huge volumes of data collected from different sources. Feature Database - handles daily operations (transactions, updates, queries). Data warehouse - support analysis, reported, and business intelligences - Database, current detailed real time data. Warhouse: historical aggregated and structured data.

companies use data warehouse, they allow business to analyze trend over time, spot patterns and forecast future outcomes. combine information from different system (sales, marketing, finance etc.) into one place.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse? Is a centralized storage system that holds vast amount of raw data in its native format structured, semi-structured and unstructured.

| Features | Data lake | ~~Data warehouse~~ |
|---|---|---|
| Data type | Raw/unprocessed | Processed, cleaned, structured |
| purpose | flexible storage for future use advanced analytics | Business intelligence, reporting historical analysis |
| Users | Data Scientists | Business Analysts managers |

companies dealing with massive, diverse data sources like IOT sensor data, social media feeds or click stream logs).

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?
Data mart is a smaller, more focused subset of a data warehouse that is tailored to meet the needs of a specific department, team, or business function -

Data warehouse - central repository that intergrates data from across the organization.

Data mart: A slice of what warehouse designed for specialized use cases. ~~Iso~~

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each. Structured - highly organized data that fits neatly into rows and columns of a database eg. customer names, phone numbers and addresses stored in a relational database.

Semi-structured data - tagged but flexible files example JSON file or XML documents where is labeled.

- Unstructured data - Raw information without predefined format or structure

example Videos, Image, audio recordings.



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6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models - is the process of creating a conceptual blue print for how data will be structured, stored and related within a system.

Data modeling is important. It helps developers, analysts and stake holders understand how data flows and connects

example without data modeling, customer info, product details and sales records might be scattered duplicated or inconsistent.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action. is the extraction of useful information from large data sets using statistical methods, machine learning and artificial intelligence. the purpose is to identify trends anomalies, and relationships that are not immediately obvious. Business use data mining to understand buying behavior, preferences and churn rist. Fraud detection.

Example

1. Retail - is used to analyze purchase histories and recommend products.
2. Bank and finance - they use data mining to detect fraudulent by spotting unusual spending patterns.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each. is software that manages databases, enabling users to interact with data through queries, updates and reports.

| DBMS | Type | Features | Use cases |
|---|---|---|---|
| oracle database | Relational multi model | advanced security support complex | large enterprises banking |
| MySQL | Relational | open-sources lightweight | web application startups - ecomerce |
| Microsoft SQL server | Relational multi model | strong integration with microsoft | corporate IT system ERP CRM |
| Postgre SQL | Relational multi model | open-source highly extensible | Research analytics applications needing flexibility |

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics. is a process used in data management to move data from one system into another. Extract pulling data out of various source systems (database, application, file API)

Transform - cleaning formatting and restructuring the data so its consistent and usable

Load - inserting the transformed data into target system like a data warehouse.

ETL is important ensures data from different sources is standardized



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10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

common data storage formats are used depending on whether the priority is readability, flexibility or performance at scale

1. CSV (comma-separated Value)
   Type - Text tabular
   Strength: Simple, human-readable, supported everywhere
   Use case - Exporting spreadsheets, quick reporting, small data sets.
2. JSON (Java Script object Notation)
   Type - Semi Structured
   Strengths - Flexible, Supports nested data, widely used in Apis
   Use case - Web applications, Api data exchange, log storage.
3. Parquet
   Type - Columnar, binary.
   Strengths, highly efficient for queries, strong compression optimized for big data.
   Use case. Cloud data warehouse (AWS, Azure, Google Big Query) large-scale analytics.
4. Avro. Row-based, binary.
   Strength: compact, fast serialization, supports schema evolution
   Use case Streaming data pipelines (Apache kafka, Hadoop ecosystems).



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## Section B: Artificial Intelligence & Machine Learning

**Research Tip**: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each. Is the field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.
Narrow AI (weak) - AI designed to perform a specific task or a limited set of tasks eg voice assistant like Siri

General AI (strong) AI with the ability to understand learn and Apply knowledge across wide range of tasks.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.
Is a branch of Artifical Intelligence where systems learn from data and improve their performance over time without being explicitly programmed.

Traditional programming - programmer writes explicit rules and logic

Machine learning - Systems learns rules from data

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work? Deep learning specialized branch of Machine learning that uses artificial neural networks - with many layers.

AI - The broad field of creating machines that can perform tasks

ML - A subset of AI where systems learn from data rather than being explicitly programmed.

DL - Subset of ML that uses multi-layered neural networks

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life. is a field AI that focuses on enabling computers to understand/interpret and generate.
Google translate - use NLP to convert between languages
Customer service chatbots - Understand questions and provide automated responses



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